



LinkedIn

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Website

Education

- University of Wisconsin – Madison**
Madison, WI
 - *Doctor of Philosophy in Biomedical Engineering* Expected Jun 2028
 - *Master of Science in Biomedical Engineering* August 2024
- University of Puerto Rico – Mayagüez**
Mayagüez, PR
 - *Bachelor of Science in Mechanical Engineering (Cum Laude)* December 2023

Experience

- University of Wisconsin – Ludwig Lab, Department of Biomedical Engineering & Neurosurgery**
Madison, WI
 - Graduate Researcher (Advisor: Dr. Kip Ludwig) June 2022 – Present
 - **Project: PS-OCT Guided Injectrode System** – Translational device development program targeting posterior tibial nerve stimulation (PTNS) for overactive bladder (OAB); focused on minimally invasive electrode delivery with quantified nerve-targeting precision.
 - * Designed and validated a PS-OCT-guided needle probe for percutaneous electrode and drug delivery.
 - * Conducted ($n = 11$) large animal (porcine) experiments; acquired and analyzed EMG and ECAP data to quantify on- and off-target stimulation as a function of implantation depth. Characterized relevant anatomy with histological images.
 - * Developed a 3D computational modeling pipeline integrating microCT, MUSE fluorescence imaging, and histological segmentation to produce subject-specific anatomical models for electrode placement guidance.
 - * Coordinated across imaging, surgical, and signal analysis teams to translate preclinical findings into iterative device design updates.
 - **Project: Electrochemical Safety Standards for Neuromodulation** – Bench and in vivo characterization of implantable electrodes to establish charge injection limits and design guidelines compliant with neuromodulation safety standards.
 - * Performed electrochemical impedance spectroscopy (EIS), voltage transient (VT) analysis, and cyclic voltammetry (CV) in order to quantify with actual neural waveforms the limitations of increased electroactive surface area through changes in geometry.
 - * Developed a quantitative framework relating electrode geometry to charge injection efficiency, identifying pulse-width-dependent transitions between geometric and electrochemical surface area-limited regimes.
- University of Wisconsin – Department of Biomedical Engineering**
Madison, WI
 - Teaching Assistant, Graduate Seminar: *Biomedical Startup Evaluation* Jan 2026 – May 2026
 - Co-instructed a graduate-level course analyzing 12 biomedical startups spanning gene editing, neuromodulation, and orthopedic devices.
 - Guided students through hazard analyses and venture capital pitch evaluations; taught interpretation of IP landscapes and SEC filings to assess commercial readiness of early-stage technologies.
 - Reinforced applied understanding of regulatory pathways, risk frameworks, and technology transfer strategy in the context of medical device development.
- Boston Scientific**
Dorado, PR
 - *Mechanical Engineering Capstone – 4Front Stake Cable to Fitting Project* January 2022 – June 2022
(Advisors: Dr. David Dooner & Jorge Soto)
 - **Project: Manufacturing Process Redesign for RELIANCE 4-FRONT™ Lead** – Process improvement initiative to reduce defect rates and increase throughput on a high-volume ISO 13485-regulated medical device manufacturing line.
 - * Designed and deployed a motorized blade fixture to replace manual cutting operations, eliminating operator-dependent variability and reducing material waste.
 - * Implemented in-line fault detection sensors to identify process deviations in real time and prevent downstream defective production batches.
 - * Estimated a 50% reduction in line dead time and an annual manufacturing cost reduction of \$422,328.

Publications & Patents

- **Patents**

- Ludwig K, Trevathan J, Chen RCH, Sanchez R. Apparatus for applying therapy to a neurovascular bundle. US Patent App. 18/828,019. 2026.

- **Journal Articles & Preprints**

- Gholston AK, **Sanchez R.**, Bigger TB, Averbek SR, Rieth L, Ludwig KA, et al. Systematic errors in DC offset measurement and mitigation at neurostimulation electrodes: causes, implications and solutions. bioRxiv. 2025. doi:10.1101/2025.12.08.691908
- Deshmukh A, Cheng K, Settell ML, Knudsen BE, LaLuzerne M, Savage J, **Sanchez R.**, et al. Identification of low-threshold off-target activation pathways during stimulation of carotid baroreceptor afferents in swine. Journal of Neural Engineering. 2026;23(3):036022.

- **Conference Abstracts**

- **Sanchez R.**, Gholston A, Knudsen B, Laluzerne M, Cheng K, Trevathan J, Settell M, Ludwig KA. Decoding on- and off-target circuit recruitment during percutaneous tibial nerve stimulation for overactive bladder. NYC Neuromodulation Conference, New York, NY. 2026.
- Deshmukh A, Cheng K, Knudsen B, Settell M, LaLuzerne M, Turk K, **Sanchez R.**, et al. ID: 323236 – Identifying off-target activation pathways for Baroreflex Activation Therapy (BAT) in swine. Neuromodulation. 2024;27(7):S140–S141.

Skills

- **Lab & Technical Skills:** Electrochemical Analysis (EIS, CV, VT), Large Animal Models, Histological Imaging, OCT Imaging, 3D Printing, CNC Machining, PCB Design, EMG/ECAP Acquisition, Risk Analysis, Fabrication Prototyping
- **Computer Skills:** COMSOL, Python, Matlab, Julia, Fusion 360, OnShape, Siemens NX, Blender, Autodesk Eagle, Synapse, Ascent, NEURON, PyFibers
- **Regulatory & Standards Knowledge:** ISO 13485 FDA-regulated preclinical protocols, technical documentation and design verification, invention disclosure and IP landscape assessment
- **Language Skills:** English (fluent), Spanish (native)